

1 Workshop Objective

Track 4 – The Induced Leak (Faraday’s Law)

Context: Capture sensitive data through inductive coupling between two parallel cables (crosstalk). *Hint:* The induced electromotive force follows $\mathcal{E} = -\frac{d\Phi}{dt}$. A rising edge of the binary signal ($0 \rightarrow 1$) generates a specific voltage pulse that reveals the bit value.

2 Theoretical Background

2.1 Faraday’s Law of Induction

Faraday’s law states that a time-varying magnetic flux through a conductive loop induces an electromotive force (EMF) proportional to the negative rate of change of the flux:

$$\boxed{\mathcal{E} = -\frac{d\Phi_B}{dt}} \quad \text{with} \quad \Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A} \quad (1)$$

The negative sign (Lenz’s law) indicates that the induced EMF opposes the change in flux. In side-channel analysis, this EMF can be measured on a “victim” cable that runs parallel to an “aggressor” cable carrying a secret digital signal.

2.2 Mutual Inductance Between Two Parallel Cables

For two parallel conductors, the magnetic flux produced by current I_1 in the aggressor cable that links the victim cable’s loop is proportional to I_1 . The constant of proportionality is the **mutual inductance** M :

$$\Phi_{2 \leftarrow 1} = M I_1 \quad (2)$$

Consequently, the voltage induced in the victim cable is:

$$V_{\text{ind}} = M \frac{dI_1}{dt} \quad (3)$$

Thus, the rapid current transitions of a digital signal (rising/falling edges) create measurable voltage pulses in the adjacent cable, leaking the bit sequence.

2.3 Geometric Calculation of M for Two Long Parallel Wires

Consider two infinitely long, parallel, thin wires of length l (with $l \gg$ separation d). The aggressor wire carries current I_1 . Using Ampère’s law, the magnetic field at a distance x from the aggressor is:

$$B(x) = \frac{\mu_0 I_1}{2\pi x} \quad (4)$$

The victim wire is placed at a distance d from the aggressor. The magnetic flux through the rectangular loop formed by the victim wire and its return path (ground) is obtained by integrating $B(x)$ over the width from the wire surface (radius r) to the separation d :

$$\Phi = \int_r^d B(x) l dx = \frac{\mu_0 I_1 l}{2\pi} \int_r^d \frac{dx}{x} = \frac{\mu_0 I_1 l}{2\pi} \ln \left(\frac{d}{r} \right) \quad (5)$$

Since $\Phi = MI_1$, we obtain the mutual inductance:

$$M = \frac{\mu_0 l}{2\pi} \ln\left(\frac{d}{r}\right) \quad (\text{for } l \gg d \gg r) \quad (6)$$

Here $\mu_0 = 4\pi \times 10^{-7}$ H/m is the vacuum permeability. This formula shows that longer cables, smaller separation, and thinner wires increase inductive crosstalk.

3 Leaking Secret Bits via Induced Pulses

3.1 From Current Edges to Voltage Spikes

A binary signal $b(t)$ (e.g., $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \dots$) produces current changes ΔI during transitions. Assuming a linear rise/fall time τ , the derivative is approximately:

$$\frac{dI_1}{dt} \approx \frac{\Delta I}{\tau} \quad (7)$$

Therefore, each rising edge induces a positive voltage spike:

$$V_{\text{rise}} \approx +M \frac{\Delta I}{\tau} \quad (8)$$

while each falling edge induces a negative spike:

$$V_{\text{fall}} \approx -M \frac{\Delta I}{\tau} \quad (9)$$

No transition yields no spike. By monitoring the victim cable with an oscilloscope or ADC, an attacker can reconstruct the entire bit sequence.

3.2 Numerical Example

Let the following parameters represent a typical embedded system:

- Cable length $l = 10 \text{ cm} = 0.1 \text{ m}$
- Separation $d = 2 \text{ mm} = 0.002 \text{ m}$
- Wire radius $r = 0.25 \text{ mm} = 0.00025 \text{ m}$
- Logic current swing $\Delta I = 10 \text{ mA}$
- Rise/fall time $\tau = 1 \text{ ns} = 1 \times 10^{-9} \text{ s}$

First compute M :

$$M = \frac{4\pi \times 10^{-7} \times 0.1}{2\pi} \ln\left(\frac{0.002}{0.00025}\right) = 2 \times 10^{-8} \ln(8) \approx 2 \times 10^{-8} \times 2.079 = 4.158 \times 10^{-8} \text{ H} = 41.58 \text{ nH} \quad (10)$$

Then the induced voltage spike:

$$V_{\text{ind}} = M \frac{\Delta I}{\tau} = 4.158 \times 10^{-8} \times \frac{0.01}{1 \times 10^{-9}} = 4.158 \times 10^{-8} \times 10^7 = 0.4158 \text{ V} = 415.8 \text{ mV} \quad (11)$$

This voltage is easily detectable with standard instrumentation, confirming the feasibility of the side-channel attack.